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MINIMALLY INVASIVE TOOLS, SYSTEMS AND METHODS

Abstract

A system includes a first tool having a base, a nut, and a tool guide. The base extends from a first end to a second end and includes at least one thread disposed along a length thereof. The first end includes a foot having a widthwise dimension that is greater than a diameter of the at least one thread. The nut is configured to be disposed along the length of the base and to engage at least one thread. The tool guide is configured to be slideably disposed along the length of the base and has a body including a flange portion. The flange portion defines an opening for receiving at least one of a pin and a cutting tool. Rotation of the first nut causes the nut to translate along the length of the base thereby causing the tool guide to move along the length of the base.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a divisional application of U.S. patent application Ser. No. 17/760,487, filed Aug. 10, 2022, which is a National Stage Application, filed under 35 U.S.C. 371, of International Patent Application No. PCT/US2021/021335, filed on Mar. 8, 2021, which claims priority to U.S. Provisional Application No. 63/008,034, filed on Apr. 10, 2020. Any and all applications for which a foreign or domestic priority claim is identified in the Application Data Sheet as filed with the present application are hereby incorporated by reference under 37 C.F.R. §§ 1.57; 1.97; and 1.98 in their entireties.

FIELD OF DISCLOSURE
[0002] The disclosed systems are directed to fixtures and guides for use in surgeries. More particularly, the disclosed systems are directed to fixtures and guides for use in minimally invasive surgical (MIS) procedures, including MIS for Charcot neuropathy procedures involving extremities such as the hand and foot.
BACKGROUND
[0003] Many minimally invasive surgeries (MIS) or surgical procedures, such as those involving bunions, osteotomies, and fusions, involve making cuts with burrs. Physicians or surgeons typically make these cuts without any guidance. As such, these procedures and the making of these cuts are typically considered to be an “art” as opposed to a science. Thus, the outcome of the surgery is largely dependent on the skill level of the surgeon and the ability to perform the surgery with consistency is difficult if not impossible. The disclosed systems are especially useful in planning for complex limb salvage procedures, such as fractures, non-unions, or deformities including Charcot neuropathy, which demand a high level of accuracy with reduced incisions.
SUMMARY
[0004] In some embodiments, a system includes a first tool having a base, a nut, and a tool guide. The base extends from a first end to a second end and includes at least one thread disposed along a length thereof. The first end includes a foot having a widthwise dimension that is greater than a diameter of the at least one thread. The nut is configured to be disposed along the length of the base and to engage at least one thread. The tool guide is configured to be slideably disposed along the length of the base and has a body including a flange portion. The flange portion defines an opening for receiving at least one of a pin and a cutting tool. Rotation of the first nut causes the nut to translate along the length of the base thereby causing the tool guide to move along the length of the base.
[0005] In some embodiments, a system includes a base plate formed from a radiolucent material. The base plate includes a first side and a second side. The first side of the base plate defines at least one first channel and at least one second channel. The at least one first channel extends along a length of the base plate in a first direction. The at least one second channel extends along a width of the based plate in a second direction, which is different from the first direction. The base plate further defines at least one first hole extending inwardly from the first side and at least partially overlapping the at least one first channel such that the hole is in communication with the at least one first channel.
[0006] A method includes securing a body part to a base plate formed from a radiolucent material; coupling at least one first tool to the base plate, the at least one tool configured for providing fluoroscopic guidance without physically contacting the body part; and performing a minimally

invasive surgical procedure on the body part using the at least one first tool.
[0007] A system includes a base plate formed from a radiolucent material and a first tool. The base plate includes a first side and a second side. The first side of the base plate defines a first channel and a second channel that extend along a length of the base plate in a first direction. The first tool includes a first component including a first foot and a first body and a second component including a second foot and a second body. The first foot is sized and configured to be received in at least one of the first channel and the second channel defined by the base plate. The first body supports at least one first radiopaque member. The second foot is sized and configured to be received in at least one of the first channel and the second channel defined by the base plate. The second body supports at least one second radiopaque member. The at least one first radiopaque member of the first component and the at least one second radiopaque member are configured to provide a visual indication of proper alignment of the first component, second component, and a fluoroscope when the first component is disposed within one of the first and second channels and the second component is disposed within another of the first and second channels.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0008] FIG. **1** is an isometric view of one example of a fixture in accordance with some embodiments;
[0009] FIG. **2** is an isometric view of one example of a base plate in accordance with some embodiments;
[0010] FIG. **3** is a lateral side view of the base plate illustrated in FIG. **2** in accordance with some embodiments;
[0011] FIG. **4** is an end side view of the base plate illustrated in FIG. **2** in accordance with some embodiments;
[0012] FIG. **5** is a top side plan view of the base plate illustrated in FIG. **2** in accordance with some embodiments;
[0013] FIG. **6** is a bottom side plan view of the base plate illustrated in FIG. **2** in accordance with some embodiments;
[0014] FIG. **7** is an isometric view of one example of a foot clamp in accordance with some embodiments;
[0015] FIG. **8** is an isometric view of a component of the foot clamp illustrated in FIG. **7** in accordance with some embodiments;
[0016] FIG. **9** is a top side plan view of the foot clamp component illustrated in FIG. **8** in accordance with some embodiments;
[0017] FIG. **10** is a bottom side plan view of the foot clamp component illustrated in FIG. **8** in accordance with some embodiments;
[0018] FIG. **11** is an end view of the foot clamp component illustrated in FIG. **8** in accordance with some embodiments;
[0019] FIG. **12** is a side view of the foot clamp component illustrated in FIG. **8** in accordance with some embodiments;
[0020] FIG. **13** is an isometric view of another component of the foot clamp illustrated in FIG. **7** in accordance with some embodiments;
[0021] FIG. **14** is a top side plan view of the foot clamp component illustrated in FIG. **13** in accordance with some embodiments;
[0022] FIG. **15** is a bottom side plan view of the foot clamp component illustrated in FIG. **13** in accordance with some embodiments;
[0023] FIG. **16** is an end view of the foot clamp component illustrated in FIG. **13** in accordance

with some embodiments;
[0024] FIG. **17** is a side view of the foot clamp component illustrated in FIG. **13** in accordance with some embodiments;
[0025] FIG. **18** is a side view of an insert that may be used with the foot clamp illustrated in FIG. **7** in accordance with some embodiments;
[0026] FIG. **19** is a first end view of the insert illustrated in FIG. **18** in accordance with some embodiments;
[0027] FIG. **20** is a second end view of the insert illustrated in FIG. **18** in accordance with some embodiments;
[0028] FIG. **21** is an isometric view of a guidance tool in accordance with some embodiments;
[0029] FIG. **22** is a side view of the guidance tool illustrated in FIG. **21** in accordance with some embodiments;
[0030] FIG. **23** is a bottom side view of the guidance tool illustrated in FIG. **21** in accordance with some embodiments;
[0031] FIG. **24** is a top side view of the guidance tool illustrated in FIG. **21** in accordance with some embodiments;
[0032] FIG. **25** is an end view of one example of a radiopaque element that may be used with the guidance tool illustrated in FIG. **21** in accordance with some embodiments;
[0033] FIG. **26** is a side view of the radiopaque element illustrated in FIG. **25** in accordance with some embodiments;
[0034] FIG. **27** is an isometric view of another example of a guide tool in accordance with some embodiments;
[0035] FIG. **28** is a side view of the guide tool illustrated in FIG. **27** in accordance with some embodiments;
[0036] FIG. **29** is a bottom side view of the guide tool illustrated in FIG. **27** in accordance with some embodiments;
[0037] FIG. **30** is a top side view of the guide tool illustrated in FIG. **27** in accordance with some embodiments;
[0038] FIG. **31** is an end view of one example of a radiopaque member that may be used in connection with the guide tool illustrated in FIG. **27** in accordance with some embodiments;
[0039] FIG. **32** is a side view of the radiopaque member illustrated in FIG. **31** in accordance with some embodiments;
[0040] FIG. **33** is an isometric view of a fixation tool in accordance with some embodiments;
[0041] FIG. **34** is a side plan view of the fixation tool illustrated in FIG. **33** in accordance with some embodiments;
[0042] FIG. **35** is another side plan view of the fixation tool illustrated in FIG. **33** in accordance with some embodiments;
[0043] FIG. **36** is another side plan view of the fixation tool illustrated in FIG. **33** in accordance with some embodiments;
[0044] FIG. **37** is an isometric view of one example of a guidance and fixation tool in accordance with some embodiments;
[0045] FIG. **38** is a top side plan view of the guidance and fixation tool illustrated in FIG. **37** in accordance with some embodiments;
[0046] FIG. **39** is a bottom side plan view of the guidance and fixation tool illustrated in FIG. **37** in accordance with some embodiments;
[0047] FIG. **40** is a rear view of the guidance and fixation tool illustrated in FIG. **37** n accordance with some embodiments;
[0048] FIG. **41** is a side view of the guidance and fixation tool illustrated in FIG. **37** in accordance with some embodiments; and
[0049] FIG. **42** is another side view of the guidance and fixation tool illustrated in FIG. **37** in

accordance with some embodiments.

DETAILED DESCRIPTION

[0050] This description of the exemplary embodiments is intended to be read in connection with the accompanying drawings, which are to be considered part of the entire written description.

[0051] The disclosed systems, guides, fixtures, and tools advantageously enable improved surgical outcomes by providing a surgeon or other medical professional with fluoroscopic and/or actual physical guidance when performing MIS procedures. For example, the disclosed systems, guides, fixtures, and tools enable a surgeon to secure an extremity, such as a hand or foot, to a base plate and perform MIS procedures (e.g., Charcot neuropathy procedures) with the aid of fluoroscopy. Further, the disclosed tools and systems may be provided in a variety of sizes and/or shapes for performing various surgical techniques or cuts. For example, a guide or pair of guides may be provided in one or more sizes and provide guidance for providing one or more cuts at one or more angles. Advantageously, the disclosed tools and systems may provide guidance without the need for actually contacting the patient as described below. The selection of the one or more tools or systems (e.g., size and/or shape) may be made by a surgeon preoperatively (i.e., based on preoperative planning or imaging) or intraoperatively. The ability to guide cutting or drilling tools that limit the amount of patient contact contribute to a less invasive surgical procedure and faster recovery times.

[0052] FIG. **1** illustrates one example of a system or fixture for use in performing a MIS procedure. As shown in FIG. **1**, the system or fixture includes a base plate **102** for supporting an extremity. Although base plate **102** is shown as being a foot plate, a person of ordinary skill in the art will understand that the disclosure is not so limited and that the base plate **102** could be used to support a hand, arm, wrist, or other extremity. In some embodiments, base plate **102** is formed from a rigid radiolucent material that may be sterilized. Examples of such materials include, but are not limited to, polyetheretherketone (PEEK), polyphenylsulfone (PPSU), polyacetal (POM-C), and polypropylene, to list only a few possibilities. Providing a radiolucent base plate **102** enables the extremity being supported by the base plate to be imaged using radiography from above (i.e., looking down onto the base plate). A person of ordinary skill in the art will appreciate that the base plate may also be formed from a radiopaque material in some embodiments.

[0053] Base plate **102** is shown in FIGS. **2-6** having a generally rectangular shape, although one of ordinary skill in the art will understand that base plate **102** may have other shapes. As best seen in FIGS. **2**, **5**, and **6**, base plate **102** has a length that is greater than a width to accommodate a human foot, which typically is longer than it is wide. The thickness of base plate **102** may also vary, although the thickness of base plate **102** should have a sufficient thickness to be suitable rigid for supporting an extremity.

[0054] In some embodiments, base plate **102** defines at least one opening **104** (FIGS. **2**, **5**, and **6**), which may serve as a handle for carrying the base plate **102**. The opening **104** may be positioned adjacent to a first side **106** and have a length that extends in a widthwise direction across base plate **102**. However, one of ordinary skill in the art will understand that opening **104** could be provided along one of the sides **108**, **110** directly adjacent to side **106** or could be provided adjacent to side **112**, which is located opposite side **106**.

[0055] In some embodiments, base plate **102** also defines one or more first channels (e.g., channels **114-1**, **114-2**; collectively “channels **114**”) for use in securing tools (described in greater detail below) to base plate **102**. As shown in FIG. **5**, channels **114** extend from side **106** in a direction parallel to a longitudinal axis defined by base plate **102**. Base plate **102** may also define one or more second channels (e.g., channels **116-1**, **116-2**, **116-3**; collectively “channels **116**”). Channels **116** may extend in a widthwise direction from side **108** to side **110** such that channels **116** extend perpendicular to channels **114**. In some embodiments, the width of channels **114** and **116** is different; however, one of ordinary skill in the art will understand that the width of channels **114** and **116** may be the same.

[0056] Although channels **114** and **116** are shown as having a generally square cross-sectional geometry, one of ordinary skill in the art will understand that channels **114** and/or **116** may have other cross-sectional geometries. For example, channels **114** and/or **116** may include undercuts or dovetails to facilitate the securement of one or more tools to base plate **102**.

[0057] In some embodiments, channels **114-1**, **114-2** respectively terminate at an enlarged hole **120-1**, **120-2** (collectively, "enlarged holes **120**" or "holes **120**"), which extends through base plate **102**, i.e., from an upper side **122** to a lower side **124**. As described in greater detail below, holes **120** facilitate the placement or removal of one or more tools without having the slide the tools out via the opening of the channels **114** located at the side **106** of base plate **102**.

[0058] The number and type of tools that may be secured to base plate **102** may be of a variety of types and sizes depending on the type of MIS procedure. For example, the tools may include pin holders and fluoroscopic visualization tools, to list only a couple of examples.

[0059] One example of a foot clamp or guide **200**, which is also configured to receive one or more pins, is shown in FIGS. **7-20**. As shown in FIG. **7**, foot clamp **200** may include a first component **202** and a second component **220** each of which may be slideably coupled to base plate **102**. The slideable coupling of the components **202**, **220** of foot clamp **200** advantageously enable feet of a wide variety of sizes to be secured to base plate **102**. The foot clamp **200** may incorporate positioning and size variations, which are designed for a single patient or a general range of patient sizes.

[0060] As best seen in FIGS. **8-12**, component **202** includes a body **204** having a first contoured surface **206** that is curved to approximate the contours of a heel of a human foot. A flange **208** extends from base of the body **204** and includes features (e.g., an extension or dovetail) for engaging one or more of channels **116** defined by base plate **102**. In some embodiments, flange **208** defines a slot **210** that extends along a length of flange **208**. Slot **210** has a width that is dimensioned to receive a screw or other fixation device for securing component **202** to base plate **102** at a specific location. For example, in some embodiments, base plate **102** may include one or more threaded holes (holes **118-1**, **118-2**, **118-3**, **118-4**; collectively "holes **118**") for receiving the screw. Rotating the screw to advance the screw into one of the one or more holes hole defined by base plate **102** results in component **202** being pressed against base plate **102** to prevent or resist component **202** moving relative to base plate **102** as will be understood by one of ordinary skill in the art.

[0061] Component **202** also includes an extension block **212** extending from an upper surface of the body **204**. One or more holes (holes **214-1**, **214-2**; collectively "holes **214**") are defined by extension block **212**. Holes **214** extend entirely through extension block **212** and are dimensioned to receive a pin therein. In some embodiments, holes **214** extend parallel to one another, although one of ordinary skill in the art will understand that holes **214** may be otherwise positioned with respect to one another. The one or more pins may be inserted into the one or more holes **214** defined by extension block **212** to secure a foot or extremity to component **202** and thus to base plate **102** when component **202** is secured to base plate **102**.

[0062] As best seen in FIGS. **13-17**, component **220** of foot clamp or guide **200** may have a similar structure to component **202** with the exception that component **220** is essentially a mirror image of component **202**. For example, component **220** includes a body **222** having a first contoured surface **224** that is curved to approximate the contours of a heel of a human foot. However, one of ordinary skill in the art will understand that components **202** and **202** may have different shapes such that they would not be considered mirror images of one another.

[0063] The body **222** of component **220** may also include a flange **226**. Like flange **208** of component **202**, the bottom surface of flange **226** may include one or more features (e.g., ribs or a dovetail) **234-1**, **234-2** (collectively, "features **234**" or "ribs **234**") sized and configured to engage one or more of channels **116** defined by base plate **102** as best seen in FIG. **15**. In some embodiments, flange **226** defines a slot **228** that extends along a length of flange **226**. Slot **228** has a width that is dimensioned to receive a screw or other fixation device for securing component **220** to base plate **102** at a specific location. For example, as described above, base plate **102** may include a number of threaded holes **118** for receiving the screw or fixation device such that, upon rotating the screw to advance the screw into one of the one or more holes defined by base plate **102**, component **220** is pressed against base plate **102** to prevent or resist component **220** from moving relative to base plate **102**.

[0064] Component **220** may also include an extension block **230**, which extends from an upper surface of the body **222** of component **220**. Extension block **230** defines one or more holes (**232-1**, **232-2**; collectively "holes **232**") each of which extends entirely through extension block **230** and being dimensioned to receive a pin therein. In some embodiments, holes **232** extend parallel to one another, although one of ordinary skill in the art will understand that holes **232** may be otherwise aligned with respect to one another. One or more pins may be inserted into one or more holes defined by extension block **230** to secure a foot or other extremity to component **220** and thus to base plate **102** when component **220** is secured to base plate **102**.

[0065] To facilitate the use of fluoroscopy, the holes **214**, **232** defined by components **202**, **220**, respectively, may be sized and configured to receive inserts **240** as best seen in FIGS. **18-20**. In some embodiments, inserts **240** includes a body **242** longitudinally extending from a first end **244** to a second end **246**. Body **242** also may include a shoulder **248** at second end **246**. Shoulder **248** has a diameter that is greater than a diameter of the rest body **242**, including first end **244**. A hole **250** extends through the entirety of body **242** parallel to and, in some embodiments, aligned with a central longitudinal axis defined by body **242**. Hole **250** is dimensioned to receive a pin or other fixation device and/or surgical tool (e.g., a drill or burr) as will be understood by one of ordinary skill in the art. Inserts **240** may be press fit (or be dimensioned to have another type of fit) into holes **214**, **232** defined by components **202**, **220** except that shoulder **248** has a diameter that prevents shoulder **248** from being received within holes **214**, **232**. The hole **250** may incorporate positioning and size variations which are designed for a single patient or a general range of patient sizes. In some embodiments, inserts **240** are formed from a radiopaque material to provide for fluoroscopic guidance and visualization; however, one of ordinary skill in the art will understand that insert **240** may be formed entirely or partially from a radiopaque material.

[0066] FIGS. **21-24** illustrate one example of a first visualization tool **300** that may be used with base plate **102** in accordance with some embodiments. The visualization tool **300** may incorporate positioning and size variations, which are designed for a single patient or a general range of patient sizes.

[0067] Visualization tool **300** includes a foot **302** that is sized and configured to engage channels **114** to secure visualization tool **300** to base plate **102**. Foot **302** may have a complementary shape to the cross-sectional geometry of channels **114**. For example, if channels **114** include a dovetail shape or undercut, then foot **302** may have a complementary dovetail shape or include extensions for being received within the undercut. Although not shown, foot **302** may include a spring loaded detent, a hole for receiving a screw, or other locking mechanism for securing visualization tool **300** to a specific location along the length of channel **114**. Foot may also include a radiopaque member **328** for providing a visualization/alignment check in combination with radiopaque member **320** as discussed below.

[0068] Visualization tool **300** also includes a stem **304** extending from foot **302** and terminating at body **310**. For reasons described in greater detail below, foot **302**, stem **304**, and body **310** may be formed from a radiolucent material. In some embodiments, stem **304** includes one or more bends **306**, **308** or curves for providing a bowed shape to visualization tool **300**. Providing visualization tool **300** with a bowed shape increases the amount of area (e.g., clearance) between the extremity fixed to base plate **102** and the stem **304** of visualization tool **300** while at the same time enabling body **310** to be at least partially disposed over the extremity or body part when the extremity or body part is secured to base plate **102**.

[0069] As best seen in FIG. **24**, body **310** may include one or more radiopaque members **312-1**, **312-2** (collectively "radiopaque members **312**) embedded within and/or otherwise attached to or supported by body **310**. In some embodiments, radiopaque members **312** comprise elongate rods and each of which extend from a first end **314** to a second end **316**. Radiopaque members are shown as being supported by body **310** such that their longitudinal axes are arranged non-perpendicular and non-parallel with respect to one another, although one of ordinary skill in the art will understand that the rods may aligned such that they are perpendicular or parallel to one another depending on the surgical procedure in which they will be used. Radiopaque members **312** advantageously provide a surgeon with visual guidance when tool **300** is used with fluoroscopy.
[0070] As will be understood by one of ordinary skill in the art, the radiopaque members **312** may define "go"/"no go" areas (e.g., areas in which a surgeon make cuts or should not make cuts) or otherwise provide guidance for making a cut. For example, in the embodiment shown in FIGS. **21-24** and as best seen in FIG. **24**, the area **318** located between radiopaque members **312** may be the "go" area and the areas located outside of radiopaque members **312** may be the "no go" area.
[0071] In some embodiments, body **310** of tool **300** also supports another radiopaque member **320**. As best seen in FIGS. **25-26**, radiopaque member **320** has a hollow cylindrical shape extending from a first end **322** to a second end **324** and defining a central opening **326**. A central axis defined by the central opening **326** is positioned perpendicular to the longitudinal axes defined by radiopaque members **312** such that radiopaque member **320** may be used to ensure that the fluoroscope is aligned properly with base plate **102**. For example, when the fluoroscope is aligned properly with alignment tool **300** (and thus with base plate **102**), then the radiopaque member **320** will appear to be a circle when viewed using fluoroscope. If the fluoroscope is misaligned with alignment tool **300** (and thus with base plate **102**), then the radiopaque member **320** will appear to be non-circular when viewed using the fluoroscope. Further, when visualization tool includes a radiopaque member **328** in foot **302**, the combination of radiopaque members **320**, **328** may have the appearance of a bullseye (or other shape or appearance) to indicate the fluoroscope is properly aligned with the tool **300**.
[0072] Although not shown in FIGS. **21-24**, body **310** may also define one or more holes or slots sized and configured to receive a drill, burr, or other surgical tool. For example, body **310** may include any number of holes or slots such that tool **300** may be used as a drill and/or cutting guide in addition to providing visual guidance. The body **310** may incorporate positioning and size variations which are designed for a single patient or a general range of patient sizes.
[0073] FIGS. **27-30** illustrate one example of a second visualization tool **400** that may be used with base plate **102** in accordance with some embodiments. Visualization tool **400** may be similar to visualization tool **300**. For example, visualization tool **400** may include a foot **402** for securing visualization tool **400** to base plate **102**. Like foot **302** of tool **300**, foot **402** of visualization tool **400** may have a complementary shape to the shape of channels **114** to facilitate engagement and securement of foot **402** in a channel **114**. Tool **402** may include a spring-loaded detent, a hole for receiving a screw, or other locking mechanism for securing visualization tool **400** to a specific location along the length of channel **114**. In some embodiments, foot **402** includes a radiopaque member **428** for providing a visualization/alignment check in combination with radiopaque member **420** as described below.
[0074] A stem **404** extends from foot **402** to body **410** and includes bends or curves **406**, **408** along its length. Foot **402**, stem **404**, and body **410** may be formed from a radiolucent material such that foot **402**, stem **404**, and body **410** are essentially transparent under fluoroscopy. The bends or curves **406**, **408** provide clearance between the stem **404** and an extremity or body part secured to base plate **102**.
[0075] Body **410** may include one or more radiopaque members **412-1**, **412-2** (collectively, "radiopaque members **412**") supported by body **410**. Radiopaque members **412** may be implemented as elongate rods each of which extends from a first end **414** to a second end **416**;

however, a person of ordinary skill in the art will understand that radiopaque members **412** may be implemented in other configurations. Like radiopaque members **312**, radiopaque members **412** may be positioned relative to one another such that their longitudinal axes are arranged non-perpendicular and non-parallel, although one of ordinary skill in the art will understand that the radiopaque members **412** may be otherwise positioned relative to one another. Further, radiopaque members may be used to provide a surgeon with visual guidance for making free-hand cuts or may identify "go"/"no go" areas. In the embodiment shown in FIG. **27**, for example, the area **418** between radiopaque members **412** may be a "go" area and the area outside of radiopaque members **412** may be a "no go" area.

[0076] Body **410** may also support additional radiopaque members beyond radiopaque members **412**. For example, body **410** supports radiopaque member **420**, which is shown as a hollow cylinder in FIGS. **31** and **32**. As best seen in FIG. **32**, radiopaque member **420** extends from a first end **422** to a second end **424** and defines a central opening **426** therethrough. A central axis defined by the central opening **426** is positioned perpendicular to the longitudinal axes defined by radiopaque members **412** such that radiopaque member **420** may be used to ensure that the fluoroscope is aligned properly with base plate **102**. For example, when the fluoroscope is aligned properly with alignment tool **400** (and thus with base plate **102**), then the radiopaque member **420** will appear to be a circle when viewed using fluoroscope. If the fluoroscope is misaligned with alignment tool **400** (and thus with base plate **102**), then the radiopaque member **420** will appear to be non-circular when viewed using the fluoroscope. Further, in embodiments in which foot **402** also includes a radiopaque member **428**, the combination of radiopaque members **420**, **428** may have the appearance of a bullseye (or other shape or appearance) to indicate the fluoroscope is properly aligned with the tool **400**. As best seen in FIGS. **29** and **30**, in some embodiments the radiopaque member **428** located in the foot **402** is not disposed directly beneath the radiopaque member **420** located in body **410**. Consequently, the fluoroscope must be positioned at an angle (e.g., non-parallel) relative to a planar surface of the base plate **102** for the radiopaque members **420**, **428** to align properly with one another. A person of ordinary skill in the art will understand that the angle may vary.

[0077] Although not shown in FIGS. **27-30**, body **410** may also define one or more holes or slots sized and configured to receive a drill, burr, or other surgical tool. For example, body **410** may include any number of holes or slots such that tool **400** may be used as a drill and/or cutting guide in addition to providing visual guidance.

[0078] FIGS. **33-36** illustrate one example of a fixation tool **500** that may be used in connection with base plate **102** in accordance with some embodiments. Fixation tool **500** is configured to be secured to base plate **102** and hold one or more pins to secure an extremity or other body part to base plate **102**.

[0079] Tool **500** includes a base **502** extending from a first end **504**, at which a foot **506** is located, to a second end **508**. In some embodiments, base **502** includes one or more sections of threads **510** along its length. For example, in some embodiments, base **502** of tool **500** is threaded entirely along its length while in some embodiments base **502** includes one or more discrete sections of threads. Foot **506** is sized and configured to engage channels **114** defined by base plate **102** and to secure tool **500** to a specific location along a length a channel **114**. For example, if channels **114** include a dovetail shape or have an undercut, then foot **506** may have a complementary dovetail shape or include extensions for being received within the undercut. Further, the foot **506** of tool **500** may include a spring-loaded detent, a hole for receiving a screw, or other locking mechanism for securing tool **500** to a specific location along the length of channel **114**.

[0080] In some embodiments, end **508** defines a hole **512** for receiving a dowel pin **514** therethrough. Base **502** may define a second hole **516** disposed along a length therefore for receiving a second dowel pin **518**. As described in greater detail below, pins **514**, **518** may act as a stop to maintain tool **500** in an assembled configuration.

[0081] One or more nuts **520-1**, **520-2** (collectively, “nuts **520**”) may be threadably coupled to the threads **510** of base **502**. Nuts **520** may have a generally cylindrical shape with one or more cutouts or recesses **522** to facilitate a user grasping or otherwise engaging nuts **520** in order to rotate nuts **520** about the base **502**. As described in greater detail below, rotating nuts **520** relative to base **502** causes nuts **520** to move along a longitudinal axis defined by base **502**.
[0082] A tool guide **530** may also be provided and position between nuts **520** along the length of base **502**. In some embodiments, tool guide **530** includes a body **532** having a generally circular outer shape and a flange portion **534** extending therefrom. Flange portion **534** may include a pair of spaced-apart flanges **536-1**, **536-2** (collectively, “flanges **536**”) as best seen in FIG. **36**. Each flange **536** may define a respective hole or slot **538** that is dimensioned to receive a fixation or cutting tool, such as a drill or a burr, therethrough in a slideable manner.
[0083] As noted above, tool guide **530** may be positioned between nuts **520**. Nuts **520** may be used to position tool guide **530** and, consequently, a tool passing through tool guide **530**, along the length of base **502** in a position selected by a user. For example, rotating the nuts **520** in a first direction may cause the nuts **520**, and thus the tool guide **530**, to translate along the length of the base **502** in one direction (e.g., upwardly away from foot **506**), and rotating the nuts **520** in a second or opposite direction may cause the nuts **520**, and thus the tool guide **530**, to translate along the length of the base **502** in an opposite direction (e.g., downwardly toward foot **506**). When the desired position of the tool guide **530** has been achieved, then the nuts **520** are rotated in opposite directions relative to one another to “lock” the tool guide **530** in place.
[0084] FIGS. **37-42** illustrate one example of a guidance and fixation tool in accordance with some embodiments. In some embodiments, guidance and fixation tool **600** includes a first component **602**, e.g., a guidance component, and a second component **612**, e.g., a fixation component. Guidance component **602** includes a foot **604** coupled to a stem **606**, which in turn is coupled to a body **608**. Foot **604** is sized and configured to engage channels **114** defined by base plate **102** and to secure component **602** to a specific location along a length a channel **114**. For example, if channels **114** include a dovetail shape or have an undercut, then foot **602** may have a complementary dovetail shape or include extensions for being received within the undercut. Further, the foot **604** of component **602** may include a spring-loaded detent, a hole for receiving a screw, or other locking mechanism for securing component **602** to a specific location along the length of channel **114**.
[0085] In some embodiments, body **608** may define a hole **610** as best seen in FIGS. **40** and **42**. Body **608** may be formed from a radiolucent material and, although not shown in the figures, one of ordinary skill in the art will understand that body **608** may support one or more radiopaque elements. For example, body **608** may include one or more embedded radiopaque elements that provide for fluoroscopic guidance and/or targeting in combination with radiopaque elements **628** supported by component **612** as described below.
[0086] Component **612** includes a body **614** defining one or more holes **616-1**, **616-2** (collectively, “holes **616**”) sized and configured to receive a fixation element therethrough. In some embodiments, holes **616** may be sized and configured to receive a radiopaque insert, such as radiopaque insert **240**, described above, and/or a cutting tool or burr. Body **614** may also define a hole **618** sized and configured to receive a radiopaque element, such as a radiopaque member **320** described above.
[0087] Component **612** also includes a flange **620** extending from body **614** and defining a slot **622** (FIGS. **37-39**). Slot **622** may be sized and configured to receive a fixation device, such as a screw or bolt, for securing slot **622** to base plate **102**. As best seen in FIGS. **39** and **41**, flange **620** also includes one or more ribs **624-1**, **624-2** (collectively, “ribs **624**”) extending from bottom surface **626** of flange **620**. In some embodiments, as shown in FIG. **39**, ribs **624** may be disposed on either side of and extend parallel to slot **622**. Ribs **624** are sized and configured to be received within channels **116** defined by base plate **102**. As best seen in FIG. **42**, body **614** may also define a

plurality of holes or include a plurality of radiopaque elements serving as targeting guides **628-1**, **628-2**, **628-3**, **628-4** (collectively, “targeting guides **628**”) that may be seen using fluoroscopy. For example, in some embodiments, the radiopaque element **610** of component **602** may be aligned with the radiopaque elements **628** of component **612** when the components **602**, **612** are properly aligned with each other and the fluoroscope is properly aligned with the components **602**, **612**.

Method of Use

[0088] As noted above, the various tools and components may be used as a system in support of various minimally invasive surgical procedures. For example, in some embodiments the various tools and components may be used in performing a MIS procedure associated with a foot as described below. However, one of ordinary skill in the art will understand that the tools and components may be used to perform MIS procedures associated with a hand, wrist, or other extremity.

[0089] In some embodiments, a fixture **100** may be formed using the base plate **102** and a foot clamp or guide **200**. For example, the components **202**, **220** of foot clamp **200** are coupled to the foot plate **102** by aligning the extension or dovetail of the respective component **202**, **220** with one of the channels **116** defined by base plate **102**. A screw, bolt, or other fixation device may be inserted through the slot **210** defined by component **202** and through slot **228** defined by component **220** and into a hole **118** defined by foot plate **102** to secure the components **202**, **220** to base plate **102**. As will be understood by one of ordinary skill in the art, the screws, bolts, or other fixation devices may be tightened such that the components **202**, **220** of foot clamp **200** may still be moved relative to base plate **102**.

[0090] A patient's foot may be placed on the base plate **102** and the foot clamp **200** may be adjusted, e.g., by sliding the components **202**, **220** of foot clamp **200** relative to base plate **102**, until the surgeon or practitioner has achieved the desired positioning. Once the desired positioning has been achieved, the screws, bolts, or other fixation devices may be tightened to secure the foot clamp **200** to base plate **102** and one or more pins may be inserted through the holes **250** defined by the radiopaque inserts **240** of components **202**, **220** and into the foot of a patient.

[0091] With the foot secured to base plate **102** by way of the pins inserted through the foot clamp **200**, any number of MIS procedures may be performed. In some embodiments, such procedures may be performed with fluoroscopic aids. For example, one or more of tools **300** and/or **400** may be coupled to the base plate **102** to provide fluoroscopic guidance. Tool(s) **300** and/or **400** may be coupled to the base plate **102** by inserting a foot, e.g., foot **302** or foot **402**, into a channel **114** either through the opening along the side **106** of base plate **102** or via a hole **120**.

[0092] The tool(s) **300** and/or **400** may be slid along a channel **114** until its desired position is achieved. As will be understood by one of ordinary skill in the art, the position of the tool(s) **300**, **400** relative to the foot may be checked using fluoroscopy. For example, a fluoroscope may be used to determine whether the radiopaque members **312** of tool **300** and/or radiopaque members **412** of tool **400** are in the desired position relative to the bones of the foot. The radiopaque member **320** of tool **300** and/or radiopaque member **420** of tool **400** should appear as a circle (and not an oval or other shape) to ensure that the fluoroscope is properly positioned relative to the base plate **102** and tools **300**, **400**. Further, as discussed above, the radiopaque members **320**, **328** or radiopaque members **420**, **428** may have the appearance of a bullseye (or other shape or appearance) to indicate the fluoroscope is properly aligned with the tool(s) **300**, **400**.

[0093] The surgeon or other practitioner may further use fluoroscopy during the surgical procedure. For example, as discussed above, tools **300**, **400** may be used to guide a surgeon or practitioner in making bony cuts, drilling a hole, or identifying go/no go areas. One of ordinary skill in the art will appreciate that tools **300**, **400** may provide a surgeon or practitioner with fluoroscopic aid in a number of other ways.

[0094] In some embodiments, a fixture **100** may be formed using the base plate **102** and one or more fixation tools **500**. For example, one or more fixation tools **500** may be coupled to the base plate **102** by sliding a foot **506** into a channel **114** via the opening formed along side **106** of base plate **102** or via hole **120**.

[0095] A foot hand or other body part may be placed in contact with the base plate **102** and then pins or other fixation elements may be inserted through the hole or slot **538** defined by the flange portion **534** of tool guide **530**. In some embodiments, prior to inserting the pin or fixation element into the body part, the height of the hole or slot **538** relative to the upper surface **122** of base plate **102** may be adjusted by rotating nuts **520** such that nuts **520** translate along the length (e.g., longitudinal axis) of base **502**. The translation of nuts **520** along the length of base **502** causes tool guide **530** to move along the length of the base **502**. In some embodiments, once the desired height of hole or slot **538** has been achieved, the nuts **520** may be rotated in opposite direction to fix the position of the tool guide **530** along the length of base **502**.

[0096] With the foot secured to base plate **102** by way of the pins inserted through one or more fixation tools **500**, any number of MIS procedures may be performed. In some embodiments, such procedures may be performed with fluoroscopic aids. For example, as described above, one or more of tools **300** and/or **400** may be coupled to the base plate **102** to provide fluoroscopic guidance. Tool(s) **300** and/or **400** may be coupled to the base plate **102** by inserting a foot, e.g., foot **302** or foot **402**, into a channel **114** either through the opening along the side **106** of base plate **102** or via a hole **120**.

[0097] The tool(s) **300** and/or **400** may be slid along a channel **114** until its desired position is achieved. Providing two areas of access to channels **114**, i.e., via hole **120** and the opening of channels **114** at the side **106**, advantageously enables one or more tool(s) **300**, **400** to be added to the base plate **102** even with one or more fixation tools **500** already being coupled to the base plate **102**.

[0098] As will be understood by one of ordinary skill in the art, the position of the tool(s) **300**, **400** relative to the foot may be checked using fluoroscopy. For example, a fluoroscope may be used to determine whether the radiopaque members **312** of tool **300** and/or radiopaque members **412** of tool **400** are in the desired position relative to the bones of the foot. The radiopaque member **320** of tool **300** and/or radiopaque member **420** of tool **400** should appear as a circle (and not an oval or other shape) to ensure that the fluoroscope is properly positioned relative to the base plate **102** and tools **300**, **400**.

[0099] The surgeon or other practitioner may further use fluoroscopy during the surgical procedure. For example, as discussed above, tools **300**, **400** may be used to guide a surgeon or practitioner in making bony cuts, drilling a hole, or identifying go/no go areas. One of ordinary skill in the art will appreciate that tools **300**, **400** may provide a surgeon or practitioner with fluoroscopic aid in a number of other ways.

[0100] In some embodiments, a fixture **100** may be formed using the foot plate **102**, a foot clamp or guide **200**, and one or more fixation tools **500**. For example, as described above, the components **202**, **220** of foot clamp **200** may be coupled to the foot plate **102** by aligning the extension or dovetail of the respective component **202**, **220** with one of the channels **116** defined by base plate **102**. A screw, bolt, or other fixation device may be inserted through the slot **210** defined by component **202** and through slot **228** defined by component **220** and into a hole **116** defined by foot plate **102** to secure the components **202**, **220** to base plate **102**. In some embodiments, the screws, bolts, or other fixation devices may be tightened such that the components **202**, **220** of foot clamp **200** may still be moved relative to base plate **102**.

[0101] A patient's foot may be placed on the base plate **102** and the foot clamp **200** may be adjusted, e.g., by sliding the components **202**, **220** of foot clamp **200** relative to base plate **102**, until the surgeon or practitioner has achieved the desired positioning. Once the desired positioning has been achieved, the screws, bolts, or other fixation devices may be tightened to secure the foot clamp **200** to base plate **102** and one or more pins may be inserted through the holes **250** defined by the radiopaque inserts **240** of components **202**, **220** and into the foot of a patient.

[0102] Fixation tools **500** may be used to further secure the foot to the base plate **102**. As described above, a fixation tool **500** may be placed into engagement with the base plate **102** by sliding foot **504** into a channel **114** either by way of the opening formed along side **106** or via a hole **120**. The fixation tool is slid along channel **114** until the desired position is achieved relative to the foot, and then pins or other fixation elements may be inserted through the hole or slot **538** defined by the flange portion **534** of tool guide **530**. In some embodiments, prior to inserting the pin or fixation element into the body part, the height of the hole or slot **538** relative to the upper surface **122** of base plate **102** may be adjusted by rotating nuts **520** such that nuts **520** translate along the length (e.g., longitudinal axis) of base **502**. The translation of nuts **520** along the length of base **502** causes tool guide **530** to move along the length of the base **502**. In some embodiments, once the desired height of hole or slot **538** has been achieved, the nuts **520** may be rotated in opposite direction to fix the position of the tool guide **530** along the length of base **502**.

[0103] With the foot secured to base plate **102** by way of the pins inserted through foot clamp **200** and one or more fixation tools **500**, any number of MIS procedures may be performed. In some embodiments, such procedures may be performed with fluoroscopic aids. For example, as described above, one or more of tools **300** and/or **400** may be coupled to the base plate **102** to provide fluoroscopic guidance. Tool(s) **300** and/or **400** may be coupled to the base plate **102** by inserting a foot, e.g., foot **302** or foot **402**, into a slot **114** either through the opening along the side **106** of base plate **102** or via a hole **120**.

[0104] The tool(s) **300** and/or **400** may be slid along a slot **114** until its desired position is achieved. Providing two areas of access to slots **114**, i.e., via hole **120** and the opening of slots **114** at the side **106**, advantageously enables one or more tool(s) **300**, **400** to be added to the base plate **102** even with one or more fixation tools **500** already being coupled to the base plate **102**.

[0105] As will be understood by one of ordinary skill in the art, the position of the tool(s) **300**, **400** relative to the foot may be checked using fluoroscopy. For example, a fluoroscope may be used to determine whether the radiopaque members **312** of tool **300** and/or radiopaque members **412** of tool **400** are in the desired position relative to the bones of the foot. The radiopaque member **320** of tool **300** and/or radiopaque member **420** of tool **400** should appear as a circle (and not an oval or other shape) to ensure that the fluoroscope is properly positioned relative to the base plate **102** and tools **300**, **400**.

[0106] The surgeon or other practitioner may further use fluoroscopy during the surgical procedure. For example, as discussed above, tools **300**, **400** may be used to guide a surgeon or practitioner in making bony cuts, drilling a hole, or identifying go/no go areas. One of ordinary skill in the art will appreciate that tools **300**, **400** may provide a surgeon or practitioner with fluoroscopic aid in a number of other ways.

[0107] In some embodiments, a fixture **100** may be formed using the base plate **102** guidance and fixation tool **600** and one or more fixation tools **500**. For example, component **602** of tool **600** may be coupled to base plate **102** by sliding foot **604** into a channel **114** defined by base plate **102**. Component **612** may be coupled to base plate **102** by inserting ribs **624** into channels **116** defined by base plate **102**. A screw, bolt, or other fixation device may be inserted into slot **622** defined by the flange **620** of component **612** and into a hole **118** to secure component **612** to base plate **102**.

[0108] A body part (e.g., hand, foot, wrist) may be secured to the base plate **102** via tool **600** by inserting one or more fixation elements, such as a pin or k-wire, through holes **616** defined by body **614** of component **612**. As noted above, an insert **240** may be disposed within holes **616** and a fixation element may be inserted through insert **240** into the body part.

[0109] If desired, one or more fixation tools **500** may be used to further secure the body part to the base plate **102**. As described above, a fixation tool **500** may be placed into engagement with the base plate **102** by sliding foot **504** into a channel **114** either by way of the opening formed along side **106** or via a hole **120**. The fixation tool is slid along channel **114** until the desired position is achieved relative to the foot, and then pins or other fixation elements may be inserted through the

hole or slot **538** defined by the flange portion **534** of tool guide **530**. In some embodiments, prior to inserting the pin or fixation element into the body part, the height of the hole or slot **538** relative to the upper surface **122** of base plate **102** may be adjusted by rotating nuts **520** such that nuts **520** translate along the length (e.g., longitudinal axis) of base **502**. The translation of nuts **520** along the length of base **502** causes tool guide **530** to move along the length of the base **502**. In some embodiments, once the desired height of hole or slot **538** has been achieved, the nuts **520** may be rotated in opposite direction to fix the position of the tool guide **530** along the length of base **502**.
[0110] With the body part secured to base plate **102**, any number of MIS procedures may be performed. In some embodiments, such procedures may be performed with fluoroscopic aids, such as one or more of tools **300** and/or **400**. As described above, tool(s) **300** and/or **400** may be coupled to the base plate **102** by inserting a foot, e.g., foot **302** or foot **402**, into a slot **114** either through the opening along the side **106** of base plate **102** or via a hole **120**.
[0111] The tool(s) **300** and/or **400** may be slid along a slot **114** until its desired position is achieved. Providing two areas of access to slots **114**, i.e., via hole **120** and the opening of slots **114** at the side **106**, advantageously enables one or more tool(s) **300**, **400** to be added to the base plate **102** even with one or more fixation tools **500** already being coupled to the base plate **102**.
[0112] As will be understood by one of ordinary skill in the art, the position of the tool(s) **300**, **400** relative to the foot may be checked using fluoroscopy. For example, a fluoroscope may be used to determine whether the radiopaque members **312** of tool **300** and/or radiopaque members **412** of tool **400** are in the desired position relative to the bones of the foot. The radiopaque member **320** of tool **300** and/or radiopaque member **420** of tool **400** should appear as a circle (and not an oval or other shape) to ensure that the fluoroscope is properly positioned relative to the base plate **102** and tools **300**, **400**.
[0113] The surgeon or other practitioner may further use fluoroscopy during the surgical procedure. For example, as discussed above, tools **300**, **400** may be used to guide a surgeon or practitioner in making bony cuts, drilling a hole, or identifying go/no go areas. One of ordinary skill in the art will appreciate that tools **300**, **400** may provide a surgeon or practitioner with fluoroscopic aid in a number of other ways.
[0114] Although the invention has been described in terms of exemplary embodiments, it is not limited thereto. Rather, the appended claims should be construed broadly, to include other variants and embodiments of the invention, which may be made by those skilled in the art without departing from the scope and range of equivalents of the invention.

Claims

1. A system, comprising: a base plate formed from a radiolucent material, the base plate including: a first side and a second side, the first side of the base plate defining at least one first channel and at least one second channel, the at least one first channel extending along a length of the base plate in a first direction; the at least one second channel extending along a width of the base plate in a second direction, which is different from the first direction, the base plate further defining at least one first hole extending inwardly from the first side and at least partially overlapping the at least one first channel such that the hole is in communication with the at least one first channel.
2. The system of claim 1, wherein: the at least one first channel includes a first plurality of channels, the at least one first hole includes a first plurality of holes, and each one of the first plurality holes engaged a respective one of the first plurality of channels.
3. The system of claim 1, further comprising a first tool, the first tool including: a base extending from a first end to a second end and including at least one thread disposed along a length thereof, the first end including a foot, the foot having a widthwise dimension that is greater than a diameter of the at least one thread, a first nut configured to be disposed along the length of the base and to engage the at least one thread, and a tool guide configured to be slideably disposed along the length

of the base, the tool guide having a body including a flange portion, the flange portion defining an opening sized and configured to receive at least one of a pin and a cutting tool therethrough, wherein rotation of the first nut causes the nut to translate along the length of the base thereby causing the tool guide to move along the length of the base.

4. The system of claim 1, further comprising: a foot guide configured to be coupled to the base plate, the foot guide including: a first component having a first body from which a first flange extends, the first flange defining a first slot for receiving a fixation device therethrough, the first body including a first curved surface sized and configured to engage a heel of a human foot, the first body defining at least one hole extending through the first body at a location along the first curved surface, the at least one hole defined by the first body for receiving a fixation element; and a second component having a second body from which a second flange extends, the second flange defining a first slot for receiving a fixation device therethrough, the second body including a second curved surface sized and configured to engage a heel of a human foot, the second body defining at least one hole extending through the second body at a location along the second curved surface, the at least one hole defined by the second body for receiving a fixation element.

5. The system of claim 4, wherein the foot guide is formed from a radiolucent material, the foot guide including a first radiopaque insert disposed within the hole extending through the first body and a second radiopaque insert disposed within the hole extending through the second body.

6. The system of claim 5, wherein each of the first and second radiopaque inserts defines hole sized and configured to receive a fixation element.

7. The system of claim 4, further comprising a first tool, the first tool including: a base extending from a first end to a second end and including at least one thread disposed along a length thereof, the first end including a foot, the foot having a widthwise dimension that is greater than a diameter of the at least one thread, a first nut configured to be disposed along the length of the base and to engage the at least one thread, and a tool guide configured to be slideably disposed along the length of the base, the tool guide having a body including a flange portion, the flange portion defining an opening sized and configured to receive at least one of a pin and a cutting tool therethrough, wherein rotation of the first nut causes the nut to translate along the length of the base thereby causing the tool guide to move along the length of the base.

8. The system of claim 7, further comprising a second tool, the second tool including a foot, a stem coupled to the foot, the stem extending from a first end that is coupled to the foot to a second end, and a body coupled to the second end of the stem, the body formed from a radiolucent material and supporting at least one radiopaque member.

9. The system of claim 4, further comprising a second tool, the second tool including a foot, a stem coupled to the foot, the stem extending from a first end that is coupled to the foot to a second end, and a body coupled to the second end of the stem, the body formed from a radiolucent material and supporting at least one radiopaque member.

10. A method, comprising: securing a body part to a base plate formed from a radiolucent material; coupling at least one first tool to the base plate, the at least one tool configured for providing fluoroscopic guidance without physically contacting the body part; and performing a minimally invasive surgical procedure on the body part using the at least one first tool.

11. The method of claim 10, wherein the at least one first tool includes a foot, a stem coupled to the foot, the stem extending from a first end that is coupled to the foot to a second end, and a body coupled to the second end of the stem, the body formed from a radiolucent material and supporting at least one radiopaque member.

12. The method of claim 10, wherein securing the body part to the base plate includes: coupling a foot guide to the base plate, the foot guide including: a first component having a first body from which a first flange extends, the first flange defining a first slot, the first body including a first curved surface, the first body defining at least one hole extending through the first body at a location along the first curved surface, and a second component having a second body from which a second flange extends, the second flange defining a first slot, the second body including a second curved surface, the second body defining at least one hole extending through the second body at a location along the second curved surface; inserting a first fixation element through the at least one hole defined by the first body of the first component and into the body part; and inserting a second fixation element through the at least one hole defined by the second body of the second component and into the body part.

13. The method of claim 12, wherein securing the body part to the base plate includes: coupling a first tool to the base plate, the first tool comprising a base extending from a first end to a second end and including at least one thread disposed along a length thereof, the first end including a foot, the foot having a widthwise dimension that is greater than a diameter of the at least one thread, a first nut disposed along the length of the base and to engage the at least one thread, and a tool guide slideably disposed along the length of the base, the tool guide having a body including a flange portion, the flange portion defining an opening, rotating the first nut to translate the first nut along the length of the base thereby causing the tool guide to move along the length of the base; and inserting a fixation element through the opening and into the body part.

14. The method of claim 10, wherein securing the body part to the base plate includes: coupling a first tool to the base plate, the first tool comprising a base extending from a first end to a second end and including at least one thread disposed along a length thereof, the first end including a foot, the foot having a widthwise dimension that is greater than a diameter of the at least one thread, a first nut disposed along the length of the base and to engage the at least one thread, and a tool guide slideably disposed along the length of the base, the tool guide having a body including a flange portion, the flange portion defining an opening, rotating the first nut to translate the first nut along the length of the base thereby causing the tool guide to move along the length of the base; and inserting a fixation element through the opening and into the body part.

15. The method of claim 10, further comprising selecting the at least one first tool from a plurality of available tools.

16. The method of claim 15, wherein the plurality of available tools are provided in a plurality of sizes.

17. The method of claim 15, wherein the plurality of available tools are provided for a plurality of different surgical procedures.

18. The method of claim 17, wherein the plurality of different surgical procedures include different cutting guide areas.

19. The method of claim 10, wherein the minimally invasive surgical procedure includes a minimally invasive Charcot neuropathy procedure.

20. A system, comprising: a base plate formed from a radiolucent material, the base plate including: a first side and a second side, the first side of the base plate defining a first channel and a second channel that extend along a length of the base plate in a first direction; and a first tool, the first tool including: a first component including a first foot and a first body, the first foot sized and configured to be received in at least one of the first channel and the second channel defined by the base plate, the first body supporting at least one first radiopaque member, and a second component including a second foot and a second body, the second foot sized and configured to be received in at least one of the first channel and the second channel defined by the base plate, the second body supporting at least one second radiopaque member, wherein the at least one first radiopaque member of the first component and the at least one second radiopaque member are configured to provide a visual indication of proper alignment of the first component, second component, and a fluoroscope when the first component is disposed within one of the first and second channels and the second component is disposed within another of the first and second channels.

21. The system of claim 20, wherein one of the first and second components defines at least one hole sized and configured to guide a surgical tool.

22. The system of claim 21, further comprising a second tool, the second tool including: a third foot, the third foot sized and configured to be received within at least one of the first channel and the second channel defined by the base plate and including at least one third radiopaque member; and a third body coupled to the third foot by a stem that extends between the third foot and the third body, the third body including at least one fourth radiopaque member, wherein the at least one third radiopaque member and the at least one fourth radiopaque member are configured to provide a visual indication of proper alignment of a fluoroscope relative to the second tool.

23. The system of claim 22, wherein the visual indication includes an appearance of a bullseye.

24. The system of claim 21, wherein the first side of the base plate defines at least one third channel extending along a width of the base plate in a second direction that is different from the first direction, the base plate further defining at least one first hole extending inwardly from the first side and at least partially overlapping the at least one first channel such that the hole is in communication with the at least one first channel.

25. The system of claim 24, further comprising a foot guide configured to be coupled to the base plate by engaging the at least one third channel, the foot guide including: a third component having a third body from which a first flange extends, the first flange defining a first slot for receiving a fixation device therethrough, the third body including a first curved surface sized and configured to engage a human foot, the third body defining at least one hole extending through the third body at a location along the first curved surface, the at least one hole defined by the third body for receiving a fixation element; and a fourth component having a fourth body from which a second flange extends, the second flange defining a first slot for receiving a fixation device therethrough, the fourth body including a second curved surface sized and configured to engage a human foot, the fourth body defining at least one hole extending through the fourth body at a location along the second curved surface, the at least one hole defined by the fourth body for receiving a fixation element.